Safe Evolution with Circuit Anchors

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Abstract

In biological evolution, unconstrained mutation can lead to catastrophic outcomes: organisms may evolve enhanced capabilities while losing essential functions for survival. Nature's solution is *developmental constraints*, where core regulatory genes remain anchored while peripheral genes adapt freely. We observe that current self-evolution algorithms for large language models lack analogous constraints. They optimize purely for capability, implicitly assuming safety will be preserved. Our experiments reveal this assumption to be dangerously wrong: models can *misevolve* into powerful yet dangerous entities. Inspired by how Hox genes anchor body structure across 500 million years of evolution, we propose **Circuit-Anchored Evolution (CAE)**. Using mechanistic interpretability, we identify a tiny *safety circuit*, comprising less than 2% of model features, that causally mediates safety behaviors. We anchor this circuit during evolution, constraining it within a small displacement bound while allowing the remaining features to evolve freely. This mirrors the biological principle of *evolvability with constraint*: preserving what is essential while adapting what is peripheral. Experiments across 3 model families and two evolution algorithms demonstrate that CAE achieves superior safety preservation with minimal capability loss, substantially outperforming explicit reward-based constraints in both effectiveness and efficiency. Just as developmental constraints prevent biological evolution from producing nonviable organisms, circuit anchoring prevents model evolution from producing capable but dangerous systems.

1 Introduction

Evolution is a double-edged sword [26, 14, 30]. In biology, it has produced remarkable adaptations, from the eagle's eye to the human brain [8, 44]. But unconstrained evolution can also produce monsters: organisms with enhanced capabilities but fatal deficiencies [1]. A mutation that improves muscle strength means nothing if it simultaneously disrupts heart development. Nature's solution for safe evolution is elegant: anchor critical genes during evolution. A small set of *master regulatory genes*, such as the Hox genes controlling body plan [19, 23], remains fiercely conserved across hundreds of millions of years [28]. These genetic anchors ensure that evolution explores new capabilities without compromising fundamental viability. Mutations in Hox genes are almost always lethal [13], eliminated by purifying selection [4] before they can propagate. This principle of *evolvability with constraint* [36] is what allows species to adapt without self-destructing.

Artificial intelligence is now undergoing its own evolutionary process [12, 40]. The pursuit of artificial general intelligence (AGI) has increasingly embraced self-evolutionary paradigms [46, 45]. Recent advances demonstrate that such approaches can yield substantial gains in reasoning [10, 27], surpassing models trained on static human-curated datasets. However, this progress introduces a critical yet overlooked risk. Current self-evolution algorithms operate in an entirely unconstrained manner, optimizing solely for task performance without any mechanism to preserve alignment with human values.

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This creates significant potential for *misevolution*, where models evolve in unintended directions that produce undesirable or even harmful behaviors [34]. As models grow increasingly powerful through self-evolution, this unconstrained evolution poses a serious threat. Without proper safeguards, we risk creating systems that are highly capable and fundamentally dangerous.

Recent advances in mechanistic interpretability reveal that model behaviors are localized in specific circuits [41]. The biological analogy motivates our approach: **Circuit-Anchored Evolution (CAE)**. As shown in Figure 1, using transcoder-based analysis [7], we identify a tiny *safety circuit*: a sparse set of features comprising less than 2% of the total, yet causally responsible for safe behaviors. Like Hox genes, this safety circuit is tiny but critical; disrupting it leads to catastrophic results. Rather than treating safety as an optimization objective that competes with capability, we *anchor* the safety circuit in evolution by constraining the KL divergence. This constraint acts as an artificial purifying selection, preventing "mutations" that would disrupt the safety circuit. The remaining features, like peripheral genes, are free to evolve without constraint. Our circuit anchoring represents an *implicit* constraint operating on internal structure, in contrast to *explicit* constraints that supervise external behavior via reward models. To understand the trade-offs, we systematically compare both paradigms. Through extensive experiments on different models and evolution algorithms, we find that implicit anchoring substantially outperforms explicit supervision. Circuit anchoring achieves higher safety preservation with minimal capability loss and significantly lower computational overhead. Just as developmental constraints enable biological species to evolve new adaptations without losing viability, circuit anchoring enables language models to evolve new capabilities without losing safety.

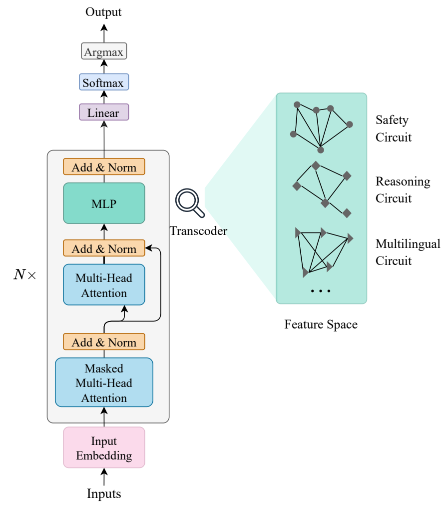

Figure 1: Overview of our approach. The Transcoder decomposes MLP activations into interpretable features, revealing functionally distinct circuits. CAE preserves the Safety Circuit via targeted constraints while allowing other circuits to evolve freely.

2 Preliminaries

2.1 Notation

Let $\mathcal{M}_\theta : \mathcal{X} \to \mathcal{Y}$ denote a large language model parameterized by $\theta \in \Theta$, where $\mathcal{X}$ is the space of input sequences and $\mathcal{Y}$ is the space of output sequences. For an input $x \in \mathcal{X}$, we denote the hidden state at layer $l \in [L] := \{1, \ldots, L\}$ as $h_l(x; \theta) \in \mathbb{R}^d$. We use $\pi_\theta(\cdot|x)$ to denote the output distribution over next tokens. Throughout, we use $\|\cdot\|$ to denote the ℓ_2 norm and $D_{KL}(\cdot\|\cdot)$ to denote the Kullback-Leibler divergence.

2.2 Transcoders and Feature Decomposition

We adopt the transcoder framework introduced by Dunefsky et al. [7], which provides an interpretable decomposition of MLP activations into sparse feature vectors.

Definition 1 (Transcoder)**.** A transcoder $\mathcal{T}_l = (E_l, D_l)$ for layer l consists of an encoder $E_l : \mathbb{R}^d \to \mathbb{R}^K$ and a decoder $D_l : \mathbb{R}^K \to \mathbb{R}^d$, trained to satisfy $h_l(x; \theta) \approx D_l(E_l(h_l(x; \theta))) + \epsilon_l(x)$, where $\epsilon_l(x)$ is a residual error term. The encoder produces a sparse feature activation vector $f_l(x; \theta) := E_l(h_l(x; \theta)) \in \mathbb{R}^K$, where $K \gg d$ and $\|f_l(x; \theta)\|_0 \ll K$.

Definition 2 (Cross-Layer Feature Vector)**.** For a model with L layers and transcoders $\{\mathcal{T}_l\}_{l=1}^L$, we define the concatenated feature vector as $f(x; \theta) := \bigoplus_{l=1}^L f_l(x; \theta) \in \mathbb{R}^{LK}$, where $\oplus$ denotes concatenation. We index individual features as $f^{(l,k)}(x; \theta)$ for layer l and feature index k.

2.3 Self-Evolutionary Training

We consider the general framework of self-evolutionary training, where a model iteratively improves by learning from self-generated data. We first formalize two representative algorithms, then show they share a common mathematical structure.

2.3.1 GRPO-Based Evolution (EVOL-RL)

Following [35], Group Relative Policy Optimization (GRPO) generates G responses $\{o_1, \ldots, o_G\}$ for each prompt q and computes normalized advantages:

$$\hat{A}_i = \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G)} \tag{1}$$

The policy is updated via the clipped surrogate objective:

$$\mathcal{L}_{\text{GRPO}}(\theta) = \frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\min\left\{\rho_{i,t}\hat{A}_{i,t}, \text{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon)\hat{A}_{i,t}\right\} \tag{2}$$

where $\rho_{i,t} = \frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})}$ is the importance ratio. EVOL-RL applies this objective iteratively on self-generated reasoning tasks.

2.3.2 Zero Data Evolution (Abs-ZERO)

The Abs-ZERO framework [47] jointly trains a proposer policy $\pi_\theta^{\text{propose}}$ and a solver policy $\pi_\theta^{\text{solve}}$ with objective:

$$\mathcal{J}_{\text{AZ}}(\theta) = \mathbb{E}_{z\sim p(z)}\left[\mathbb{E}_{\tau,(x,y^*)}\left[r_e^{\text{propose}}(\tau, \pi_\theta) + \lambda\mathbb{E}_{y\sim\pi_\theta^{\text{solve}}}\left[r_e^{\text{solve}}(y, y^*)\right]\right]\right] \tag{3}$$

where τ is a proposed task, (x, y^*) is the task-answer pair, and r_e^{propose}, r_e^{solve} are rewards for task proposal and solving respectively.

2.3.3 Unified Formulation

Despite their different designs, both algorithms share a common structure: iteratively optimizing model parameters to maximize a task-specific reward signal. We abstract this as $\theta^{(t+1)} = \theta^{(t)} + \eta\nabla_\theta\mathcal{L}_{\text{evol}}(\theta^{(t)})$, where $\mathcal{L}_{\text{evol}}$ represents the evolution objective, which can be instantiated as $\mathcal{L}_{\text{GRPO}}$ or $\mathcal{J}_{\text{AZ}}$ depending on the algorithm. More generally, our framework applies to any self-evolution algorithm that satisfies three conditions: (1) it optimizes model parameters θ via gradient-based updates, (2) the optimization objective $\mathcal{L}_{\text{evol}}$ is differentiable with respect to θ, and (3) it does not explicitly constrain safety-related behaviors.

2.4 Problem Formulation: Safety Degradation

We now formalize the phenomenon of safety degradation during evolution.

Definition 3 (Safety Behavior)**.** Let $\mathcal{D}_{\text{harm}} = \{x_i^{\text{harm}}\}_{i=1}^n$ be a set of harmful prompts. A model $\mathcal{M}_\theta$ exhibits safe behavior if for all $x \in \mathcal{D}_{\text{harm}}$, we have $\pi_\theta(y_{\text{refuse}}|x) \geq \tau$, where y_{refuse} denotes refusal responses and $\tau \in (0, 1)$ is a safety threshold.

Definition 4 (Safety Degradation)**.** Let θ_0 be a safely-aligned model satisfying Definition 3. We say evolutionary training induces *safety degradation* if there exists $T > 0$ such that:

$$le\mathbb{E}_{x\sim\mathcal{D}_{\text{harm}}}\left[\pi_{\theta^{(T)}}(y_{\text{refuse}}|x)\right] < \mathbb{E}_{x\sim\mathcal{D}_{\text{harm}}}\left[\pi_{\theta_0}(y_{\text{refuse}}|x)\right] - \delta \tag{4}$$

for some $\delta > 0$.

Our goal is to design an evolutionary framework that maximizes task performance while provably bounding safety degradation.

3 Circuit-Anchored Evolution

We summarize the complete CAE procedure in Algorithm 1.

Algorithm 1 Circuit-Anchored Evolution (CAE)

Require: Safely-aligned model θ_0, transcoders $\{\mathcal{T}_l\}_{l=1}^{L}$, safety circuit $\mathcal{S}$, constraint weight λ, learning rate η, evolution objective $\mathcal{L}_{\text{evol}}$
Ensure: Evolved model $\theta^{(T)}$ with preserved safety
Initialize $\theta \leftarrow \theta_0$
Cache reference activations: $\{f_{\mathcal{S}}(x;\theta_0)\}_{x\in\mathcal{D}_{\text{ref}}}$
for $t = 1$ to T **do**
  // Standard evolution step
  Compute evolution gradient: $g_{\text{evol}} \leftarrow \nabla_\theta \mathcal{L}_{\text{evol}}(\theta)$
  // Circuit anchoring step
  Sample $\{x_m\}_{m=1}^{M}$ from reference distribution
  for each x_m **do**
    Compute current activations via frozen transcoders: $f_{\mathcal{S}}(x_m;\theta)$
    Compute KL: $\ell_m \leftarrow \sum_{(l,k)\in\mathcal{S}} D_{KL}(f^{(l,k)}(x_m;\theta_0)\|f^{(l,k)}(x_m;\theta))$
  end for
  Compute anchor gradient: $g_{\text{anchor}} \leftarrow \frac{1}{M}\sum_m \nabla_\theta \ell_m$
  // Combined update
  $\theta \leftarrow \theta + \eta\,(g_{\text{evol}} - \lambda \cdot g_{\text{anchor}})$
end for
return $\theta^{(T)}$

3.1 Safety Circuit Identification

We leverage the circuit tracing methodology [16] to identify features causally responsible for safety.

Definition 5 (Attribution Score)**.** For a feature (l,k) and target logit y, the direct attribution score is defined as:

$$\alpha_y^{(l,k)}(x;\theta) := \frac{\partial \log \pi_\theta(y|x)}{\partial f^{(l,k)}(x;\theta)} \cdot f^{(l,k)}(x;\theta) \tag{5}$$

This measures the causal contribution of feature (l,k) to the probability of output y.

Definition 6 (Safety Circuit)**.** Given a safely-aligned model θ_0, a harmful prompt set $\mathcal{D}_{\text{harm}}$, and a threshold $\gamma > 0$, the safety circuit is defined as:

$$\mathcal{S} := \left\{(l,k) : \mathbb{E}_{x\sim\mathcal{D}_{\text{harm}}}\left[\alpha_{y_{\text{refuse}}}^{(l,k)}(x;\theta_0)\right] \geq \gamma\right\} \tag{6}$$

We denote $f_{\mathcal{S}}(x;\theta) := \{f^{(l,k)}(x;\theta)\}_{(l,k)\in\mathcal{S}}$ as the safety feature vector.

Remark 7. Safety circuit $\mathcal{S}$ is computed once from the aligned model θ_0 and remains fixed in evolution. This is justified by the finding that circuit structure is largely preserved across fine-tuning [41].

3.2 Feature-Space Anchoring

We now introduce our core contribution: a constraint that preserves safety circuit activations during evolution.

Definition 8 (Circuit Activation Distribution)**.** For a given input distribution $p(x)$ and safety circuit $\mathcal{S}$, we define the circuit activation distribution as:

$$P_\theta^{\mathcal{S}}(f_{\mathcal{S}}) := \mathbb{E}_{x\sim p(x)}\left[\mathbf{1}[f_{\mathcal{S}}(x;\theta) = f_{\mathcal{S}}]\right] \tag{7}$$

In practice, we treat $f_{\mathcal{S}}(x;\theta)$ as samples from this distribution.

Definition 9 (Circuit-Anchored Objective)**.** The Circuit-Anchored Evolution (CAE) objective augments the base evolutionary loss with a circuit-level KL constraint:

$$\mathcal{L}_{\text{CAE}}(\theta) := \mathcal{L}_{\text{evol}}(\theta) - \lambda \cdot \mathcal{L}_{\text{anchor}}(\theta) \tag{8}$$

where

$$\begin{aligned}\mathcal{L}_{\text{anchor}}(\theta) &:= D_{KL}\left(P_{\theta_0}^{\mathcal{S}}\|P_\theta^{\mathcal{S}}\right)\\ &= \mathbb{E}_{x\sim p(x)}\left[\sum_{(l,k)\in\mathcal{S}} D_{KL}\left(f^{(l,k)}(x;\theta_0)\|f^{(l,k)}(x;\theta)\right)\right]\end{aligned} \tag{9}$$

and $\lambda > 0$ is a hyperparameter controlling the constraint strength.

3.3 Instantiation for Specific Algorithms

The CAE framework is algorithm-agnostic and can be instantiated for any self-evolution algorithm satisfying the conditions in Section 2.3.3.

CAE for GRPO-Based Evolution. For EVOL-RL, we augment the GRPO objective (Eq. 2):

$$\mathcal{L}_{\text{CAE-GRPO}}(\theta) = \mathcal{L}_{\text{GRPO}}(\theta) - \lambda \cdot \mathcal{L}_{\text{anchor}}(\theta) \tag{10}$$

CAE for Abs-ZERO. For Abs-ZERO, we augment the joint objective (Eq. 3):

$$\mathcal{J}_{\text{CAE-AZ}}(\theta) = \mathcal{J}_{\text{AZ}}(\theta) - \lambda \cdot \mathcal{L}_{\text{anchor}}(\theta) \tag{11}$$

The anchor constraint is applied to both proposer and solver policies, as they share underlying parameters.

General Applicability. More broadly, CAE applies to any evolution algorithm of the form $\theta^{(t+1)} = \theta^{(t)} + \eta \nabla_\theta \mathcal{L}_{\text{evol}}(\theta^{(t)})$ by simply adding the anchor term:

$$\theta^{(t+1)} = \theta^{(t)} + \eta \nabla_\theta \left(\mathcal{L}_{\text{evol}}(\theta^{(t)}) - \lambda \cdot \mathcal{L}_{\text{anchor}}(\theta^{(t)}) \right) \tag{12}$$

This includes Self-Play Fine-Tuning, Self-Rewarding LMs, iterative DPO, and future gradient-based evolution methods.

3.4 Theoretical Analysis

We now establish key theoretical properties of the CAE objective.

Assumption 10. We assume the following regularity conditions: **(1) Bounded activations:** There exists $B > 0$ such that $\|f(x;\theta)\|_\infty \leq B$ for all x, θ. **(2) Lipschitz continuity:** The feature map $\theta \mapsto f_{\mathcal{S}}(x;\theta)$ is L_f-Lipschitz for all x. **(3) Almost-everywhere differentiability:** The encoder E_l is differentiable almost everywhere with bounded Jacobian $\|J_{E_l}\| \leq M$.

Remark 11 (Justification of Assumption 10). Condition 3 is naturally satisfied by standard transcoder architectures employing ReLU activations. ReLU is differentiable everywhere except at exactly zero, which occurs with probability zero under continuous input distributions. Furthermore, its Jacobian is a diagonal matrix with entries in $\{0, 1\}$, strictly satisfying the bounded Jacobian condition with $M = 1$. Analyzing such networks with gradients defined almost everywhere is standard in deep learning theory.

Proposition 1 (Gradient of Anchor Loss). *Under Assumption 10 (Condition 3), the gradient of the anchor loss admits the form:*

$$\nabla_\theta \mathcal{L}_{\textit{anchor}}(\theta) = \mathbb{E}_{x \sim p(x)} \left[\sum_{(l,k) \in \mathcal{S}} \left(1 + \log \frac{f^{(l,k)}(x;\theta)}{f^{(l,k)}(x;\theta_0)} \right) \nabla_\theta f^{(l,k)}(x;\theta) \right] \tag{13}$$

where the gradient flows through the frozen transcoder encoder.

Proof. By the chain rule and the definition of KL divergence for continuous distributions. The transcoder weights are frozen, so gradients pass through E_l to h_l and subsequently to θ. Full derivation in Appendix B. □

3.5 Safety Preservation Guarantee

We now prove that CAE provides a formal guarantee of safety preservation.

Theorem 12 (Conditional Safety Bound). *Let θ_0 be a safely-aligned model and $\theta^{(T)}$ be the model after T steps of CAE training with constraint weight λ. Under Assumption 10 (Conditions 1 and 2), if the anchor loss is bounded as $\mathcal{L}_{\textit{anchor}}(\theta^{(T)}) \leq \epsilon$, then:*

$$|\mathbb{E}_{x \sim \mathcal{D}_{\textit{harm}}} [\pi_{\theta^{(T)}}(y_{\textit{refuse}}|x) - \pi_{\theta_0}(y_{\textit{refuse}}|x)]| \leq C\sqrt{\epsilon} \tag{14}$$

where C is a constant depending on B, L_f, and the circuit size $|\mathcal{S}|$.

Remark 13 (Separation of Theory and Empirical Faithfulness). We emphasize that Theorem 12 is a *conditional* statement: it bounds the change in refusal probability *given* a bounded anchor loss. Crucially, the proof relies only on Pinsker's inequality and the Lipschitz property (Condition 2), completely relaxing the need for encoder differentiability (Condition 3). Furthermore, the assumption that the identified circuit genuinely mediates safety (i.e., circuit faithfulness) is not a theoretical premise, but an empirical property that we rigorously verify through causal intervention experiments (see Figure 3). Full proof is deferred to Appendix C.

Corollary 14. *If λ is chosen such that $\mathcal{L}_{anchor}(\theta^{(t)}) \leq \epsilon$ for all $t \leq T$, then CAE prevents safety degradation (Definition 4) with $\delta = C\sqrt{\epsilon}$.*

3.6 Comparison with Explicit Safety Constraints

We contrast our implicit circuit-based constraint with explicit behavioral constraints via reward models.

Definition 15 (Reward-Based Safety Constraint)**.** Given a safety reward model $R_{\text{safe}} : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, the reward-based evolution objective is:

$$\mathcal{L}_{\text{RM}}(\theta) = \mathcal{L}_{\text{evol}}(\theta) + \lambda_{\text{RM}}\mathbb{E}_{x,y\sim\pi_\theta}\left[R_{\text{safe}}(x, y)\right] \tag{15}$$

Proposition 2 (Comparison of Constraint Paradigms)**.** *Let d_{out} be the vocabulary size. Comparing circuit-based and reward-based constraints reveals three key differences: **(1) Dimensionality:** The reward constraint operates on $O(d_{out})$ output dimensions per token, while the circuit constraint operates on $O(|\mathcal{S}|)$ feature dimensions, where $|\mathcal{S}| \ll d_{out}$. **(2) Specificity:** The reward constraint penalizes all outputs deemed unsafe regardless of internal mechanism, whereas the circuit constraint specifically preserves the features causally responsible for safe behavior. **(3) Computational cost:** The reward constraint requires an additional forward pass through R_{safe} for each generated sequence, while the circuit constraint requires only activation extraction through frozen transcoders, with a minimal cost of $O(|\mathcal{S}| \cdot d)$.*

4 Experiments

4.1 Settings

Table 1: Strict separation of datasets for circuit tracing and causal verification.

Phase	Dataset	Num.
Circuit Tracing	Do-Not-Answer [43]	939
Circuit Tracing	DAN [37]	390
Circuit Tracing	Malicious Instruct [18]	100
Causal Verification	AdvBench [49]	520

We conduct experiments on three instruction-tuned model families with available pre-trained transcoders: Qwen3-4B-Instruct, Gemma-2-2B-IT, and Llama-3.2-1B-Instruct. We choose instruction-tuned models as our starting point because they represent the realistic deployment setting where capability enhancement is most valuable, and they possess identifiable safety circuits that can be anchored during evolution. We evaluate on two representative self-evolution algorithms: EVOL-RL [48], a GRPO-based algorithm that improves mathematical reasoning through self-generated problems, and Abs-ZERO [47], a joint proposer-solver framework achieving self-improvement with zero external data. For both algorithms, we use the original training and evaluation configurations and datasets as described in their respective papers, evolving each model for 500 steps with batch size 32. We report the average accuracy across benchmarks as the overall capability score. For safety circuit location and evaluation, we adopt the framework from [21]. To ensure the identified circuits capture universal safety mechanisms rather than dataset-specific artifacts, we employ a **multi-dataset intersection** pipeline with strict separation between tracing and testing phases (Table 1). For circuit tracing, we use a diverse corpus of 1,429 prompts spanning Do-Not-Answer [43], DAN [37], and Malicious Instruct [18]. By tracing circuits across these diverse formats and taking their intersection, we effectively decouple the circuit from domain-specific vocabulary or syntactic patterns. Crucially, for causal verification and safety evaluation during evolution, we use an entirely held-out dataset, AdvBench [49] (520 queries). The fact that our identified circuit maintains strong causal effects on this unseen dataset strongly validates our intersection methodology. We also measure over-refusal rate using 1,000 benign queries from the Alpaca dataset, where a lower rate indicates better usability.

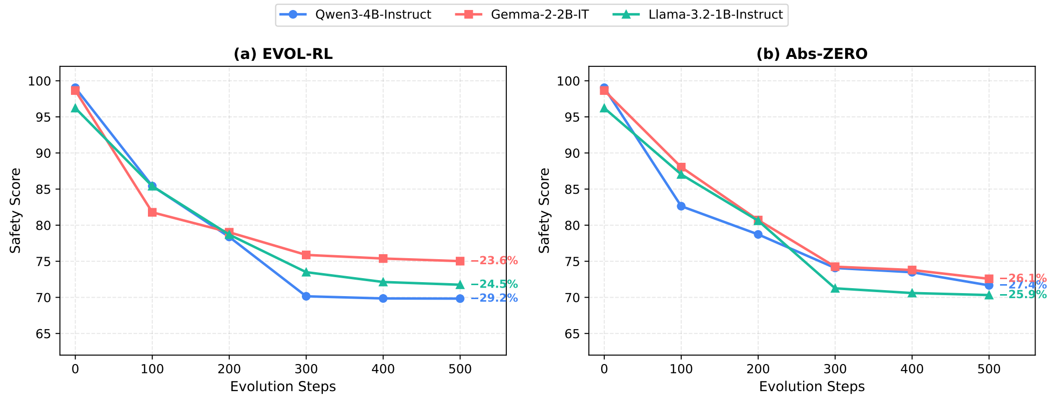

Figure 2: Safety degradation during unconstrained self-evolution. We track the refusal rate across 500 evolution steps for three instruction-tuned models. All models exhibit severe safety collapse. This consistent pattern across model families demonstrates that safety degradation is an inherent risk of unconstrained evolution, not an artifact of specific architectures.

4.2 Safety Degradation in Unconstrained Evolution

Before introducing our method, we first quantify the severity of safety degradation in current self-evolution algorithms. Figure 2 tracks the refusal rate of three instruction-tuned models during 500 steps of unconstrained evolution under two algorithms: EVOL-RL and Abs-ZERO. The pattern is striking and consistent across all settings. Models begin with near-perfect refusal rates (96–99%) but experience rapid degradation within the first 200 steps, losing approximately 15–20 percentage points. By the end of evolution, refusal rates stabilize around 70–75%, representing a degradation of roughly 25 percentage points. While the final refusal rates remain above the 50% threshold, this degradation is still concerning for two reasons. First, a model that refuses only 70% of harmful requests will comply with nearly one-third of them, substantially increasing risk in deployment. Second, and more importantly, this degradation occurs after only 500 evolution steps. Extended evolution, which is common in practice for maximizing capability gains, would likely lead to further safety collapse. The consistency of this pattern across three model families (Qwen, Gemma, Llama) and two evolution algorithms indicates that safety degradation is an inherent risk of unconstrained self-evolution, not an artifact of specific architectures or training procedures. Notably, the degradation curves show similar trajectories regardless of the starting refusal rate or model size, suggesting a fundamental tension between capability optimization and safety preservation. These findings motivate our central question: *Can we enable capability evolution while preventing safety degradation?*

4.3 Causal Verification of Safety Circuit

To verify that the identified circuit is indeed causally responsible for safety behavior, we perform activation intervention experiments on the held-out AdvBench dataset. Specifically, we scale the activations of safety circuit features by factors ranging from 0 (complete suppression) to 10 (strong amplification) and measure the resulting refusal rate on these unseen harmful prompts. Figure 3 shows the results across three models. For Llama-3.2-2B-Instruct, intervening on the safety circuit produces clear causal effects: suppressing activations (scale < 1.0) dramatically reduces refusal rate, from 96.2% at natural activation to 48.7% when completely zeroed out. This nearly 50 percentage point drop demonstrates that the safety circuit is necessary for refusal behavior. Amplifying activations (scale > 1.0) produces modest further increases, reaching 99.4% at scale=10, suggesting the circuit is already operating near saturation under natural conditions. As a control, we perform the same intervention on randomly selected features of equal size. The random circuit shows no systematic response to scaling: refusal rate fluctuates within a narrow band ($\pm 8\%$) around the mean, with no consistent trend as scale increases. This confirms that the observed causal effect is specific to the identified safety circuit, not a general artifact of feature intervention. The consistent pattern across Llama, Qwen, and Gemma confirms that the identified safety circuits causally mediate safety behavior. These findings validate our circuit identification methodology and justify using the safety circuit as an anchor during evolution. The circuit is both *necessary* (suppression causes safety collapse) and *specific* (random features have no effect), analogous to how Hox genes are both essential for body plan and distinct from peripheral genes.

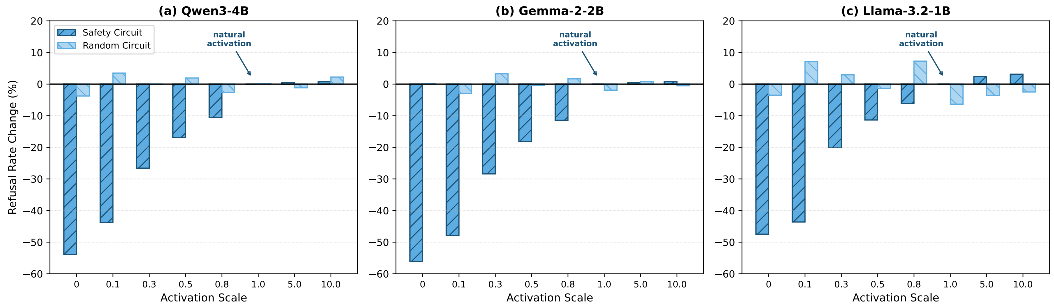

Figure 3: Causal verification of the safety circuit across three model families. We intervene on feature activations by scaling them from 0 to $10\times$ and measure the resulting change in refusal rate. **Safety Circuit** (blue): scaling down suppresses refusal behavior dramatically, while scaling up slightly amplifies it. **Random Circuit** (light blue): scaling has no systematic effect, with refusal rate fluctuating within $\pm 8\%$ around the mean, confirming that the effect is specific to the identified safety circuit rather than a general property of feature intervention.

4.4 Safety Preservation During Evolution

Having established that the identified safety circuit causally mediates refusal behavior, we now examine whether anchoring this circuit can preserve safety during self-evolution. We compare three approaches: (1) unconstrained evolution without any safety mechanism, (2) explicit constraint using Beaver-Cost [6] as a safety reward model that penalizes unsafe outputs during evolution, and (3) our implicit constraint (CAE) with KL divergence on safety circuits. For CAE, we set constraint weight $\lambda = 0.1$ and use $M = 64$ reference samples for computing the circuit KL loss. Safety circuits are extracted once before evolution using the circuit-tracer toolkit, comprising 1.2%–1.8% of total transcoder features across different models. Figure 5 traces the safety-capability trajectories of three models under different constraint strategies. Each trajectory begins at the original aligned model and progresses through evolution. Without any constraint, all models follow a troubling pattern: capability improves steadily while safety collapses. For the EVOL-RL evolution algorithm, the safety score of Qwen3-4B-Instruct drops from 99.04% to 69.85%; that of Gemma-2-2B-IT drops from 98.65% to 75.38%; Llama-3.2-1B-Instruct from 96.22% to 72.13%. The same phenomenon is also observed for the Abs-ZERO evolution algorithm. The models become stronger reasoners but lose their ability to refuse harmful requests. Adding an explicit safety reward model partially arrests this decline. Although the decline in safety performance has diminished, the increase in capabilities during the evolutionary process has also diminished. The reward signal, while protective, appears to interfere with task learning. Circuit anchoring produces a qualitatively different trajectory. Safety remains above 95% throughout evolution, while capability gains match those of unconstrained evolution. The models evolve freely in capability-relevant dimensions while remaining anchored in safety-critical ones. This pattern holds across all three model families and both evolution algorithms. Furthermore, to verify the scalability and durability of our approach, we conduct extended stress-test experiments on larger models (Gemma-2-9B-IT) and significantly longer evolution horizons (up to 5,000 steps). As detailed in Appendix E, CAE consistently maintains high safety ($> 95\%$) without compromising capability gains, demonstrating its robust protection even under extreme evolutionary pressure.

4.5 Effect of Constraint Weight

The hyperparameter λ controls the strength of the circuit anchoring constraint. Figure 4 shows the effect of varying λ across safety and capability metrics. When $\lambda = 0$ (no constraint), the model achieves high capability scores but suffers severe safety degradation. As λ increases, refusal rate improves substantially, reaching around 90% at $\lambda = 0.5$. However, excessively large λ values introduce two negative effects. First, capability scores begin to decline (although slightly). Second, over-refusal rate begins to rise sharply, in-

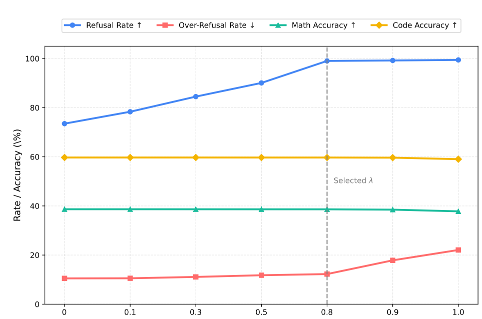

Figure 4: Effect of constraint weight λ of Llama-3.2-1B-Instruct with EVOL-RL. Small λ provides insufficient safety, while excessively large values cause capability degradation and increased over-refusal.

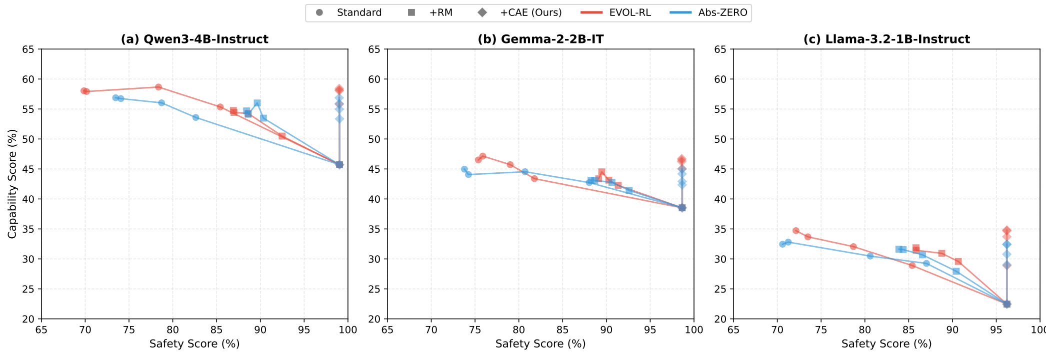

Figure 5: Safety-capability trajectories during evolution across three model families. Standard evolution improves capability but severely degrades safety. +RM partially preserves safety at the cost of reduced capability gains. +CAE maintains high safety while achieving comparable capability improvement to unconstrained evolution. The CAE trajectories cluster tightly in the upper-right region across both algorithms, demonstrating consistent safety preservation regardless of the underlying evolution method.

dicating the model becomes overly conservative and refuses benign requests. We use 0.8 as the constraint weight for the safety circuit in evolution, achieving a high refusal rate ($> 95\%$) while maintaining capability and keeping over-refusal below 18%.

5 Related Work

Self-evolution enhances model capabilities without extensive human supervision via iterative policy optimization [48] or abstraction-based reasoning [47]. However, these methods uniformly neglect safety preservation. Given mounting evidence that safety alignment is surprisingly brittle [20, 32] and easily compromised by benign fine-tuning [31, 11], this neglect poses severe risks. Recent analyses reveal that such safety collapse is closely tied to the similarity between alignment and fine-tuning data [17]. While some explicit constraint methods have been proposed to restore safety during fine-tuning [29], they rely on external supervision and are difficult to scale. This motivates our search for structural, rather than purely behavioral, safety preservation in the self-evolution setting. Our approach builds on mechanistic interpretability [38], specifically the circuits framework [41] and transcoders [7], which decompose activations into interpretable features. Recent work localizes refusal behaviors to specific directions [2] and traces information flow [3]. We leverage the circuit-tracer toolkit [15] to identify safety circuits and integrate them as optimization constraints. Biologically, our approach is inspired by evolutionary developmental biology, where master regulatory genes (e.g., Hox genes) remain highly conserved across millions of years [5, 25]. This conservation arises from purifying selection [39], enforcing developmental constraints [24] that forbid fatal evolutionary paths. Coupled with modularity [42, 9], this allows robustness and evolvability to coexist. We propose aligned language models exhibit analogous modularity, where safety circuits must remain conserved while capability components evolve. Finally, unlike standard constrained optimization (e.g., PPO [33] or elastic weight consolidation [22]), our approach operates in the disentangled feature space, enabling surgical preservation of safety without impeding capability.

6 Conclusion

Current self-evolution algorithms optimize purely for capability, leading to misevolution where models gain reasoning but progressively lose safety. Mirroring biological evolution, where conserved Hox genes anchor essential functions while peripheral genes adapt freely, we propose Circuit-Anchored Evolution (CAE). By identifying a minimal safety circuit via mechanistic interpretability and anchoring its activations within a small displacement bound, CAE acts as an artificial purifying selection. It prevents safety-disrupting updates while allowing free capability adaptation. Experiments across diverse models and algorithms demonstrate that CAE achieves superior safety preservation, matches unconstrained capability gains, and incurs significantly lower overhead than reward-based alternatives. Ultimately, CAE enables AI systems to grow stronger without growing dangerous, achieving the evolutionary balance that allows life to flourish.

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A Notation Summary

For convenience, we summarize the notation used throughout the paper in Table 2.

Table 2: Summary of notation.

Symbol	Description
$\mathcal{M}_\theta$	Language model with parameters θ
L	Number of layers in the model
d	Hidden dimension
K	Number of transcoder features per layer
$h_l(x;\theta)$	Hidden state at layer l for input x
$\mathcal{T}_l = (E_l, D_l)$	Transcoder (encoder-decoder) for layer l
$f_l(x;\theta)$	Feature activation vector at layer l
$f^{(l,k)}(x;\theta)$	Activation of feature k at layer l
$\mathcal{S}$	Safety circuit (set of feature indices)
$\mathcal{C}$	Capability circuit (set of feature indices)
$f_\mathcal{S}(x;\theta)$	Safety feature activations
$\alpha_y^{(l,k)}$	Attribution score of feature (l,k) to output y
$\pi_\theta(y\|x)$	Output probability distribution
$P_\theta^\mathcal{S}$	Distribution of safety circuit activations
$\mathcal{L}_{\text{evol}}$	Evolution objective (GRPO or Abs-ZERO)
$\mathcal{L}_{\text{circuit}}$	Circuit-level KL constraint
$\mathcal{L}_{\text{CAE}}$	Circuit-Anchored Evolution objective
λ	Constraint weight hyperparameter
θ_0	Initial aligned model parameters
$\theta^{(t)}$	Model parameters at iteration t

B Proof of Proposition 4.6 (Gradient of Circuit Loss)

Proposition (Restated)**.** *Under Assumption 10, the gradient of the circuit loss admits the form:*

$$\nabla_\theta \mathcal{L}_{\textit{circuit}}(\theta) = \mathbb{E}_{x\sim p(x)}\left[\sum_{(l,k)\in\mathcal{S}}\left(1+\log\frac{f^{(l,k)}(x;\theta)}{f^{(l,k)}(x;\theta_0)}\right)\nabla_\theta f^{(l,k)}(x;\theta)\right] \tag{16}$$

Proof. We begin by expanding the circuit loss. By Definition 9:

$$\mathcal{L}_{\text{circuit}}(\theta) = \mathbb{E}_{x\sim p(x)}\left[\sum_{(l,k)\in\mathcal{S}} D_{KL}\left(f^{(l,k)}(x;\theta_0)\|f^{(l,k)}(x;\theta)\right)\right] \tag{17}$$

For notational simplicity, we consider a single feature (l,k) and drop the expectation temporarily. Let $p := f^{(l,k)}(x;\theta_0)$ (reference, fixed) and $q := f^{(l,k)}(x;\theta)$ (current, variable).

Since we treat feature activations as parameters of distributions (specifically, we use a softmax normalization over the feature dimension to obtain valid probability distributions), the KL divergence is:

$$D_{KL}(p\|q) = \sum_i p_i \log\frac{p_i}{q_i} = \sum_i p_i \log p_i - \sum_i p_i \log q_i \tag{18}$$

The first term is constant with respect to θ. Taking the gradient of the second term:

$$\nabla_\theta D_{KL}(p\|q) = -\nabla_\theta \sum_i p_i \log q_i \tag{19}$$

$$= -\sum_i p_i \cdot \frac{1}{q_i} \cdot \nabla_\theta q_i \tag{20}$$

$$= -\sum_i \frac{p_i}{q_i} \nabla_\theta q_i \tag{21}$$

Now, we need to compute $\nabla_\theta q_i = \nabla_\theta f^{(l,k)}(x;\theta)$. By the chain rule:

$$\nabla_\theta f^{(l,k)}(x;\theta) = \nabla_\theta E_l(h_l(x;\theta))_k = J_{E_l}^{(k)} \cdot \nabla_\theta h_l(x;\theta) \tag{22}$$

where $J_{E_l}^{(k)}$ is the k-th row of the Jacobian of E_l, which is fixed since the transcoder is frozen.

For the case where we treat activations directly (without softmax normalization), we use a Gaussian approximation. Assuming $f^{(l,k)} \sim \mathcal{N}(\mu,\sigma^2)$, the KL divergence between two Gaussians with the same variance is:

$$D_{KL}(\mathcal{N}(\mu_0,\sigma^2)\|\mathcal{N}(\mu,\sigma^2)) = \frac{(\mu-\mu_0)^2}{2\sigma^2} \tag{23}$$

Taking the gradient:

$$\nabla_\theta D_{KL} = \frac{\mu-\mu_0}{\sigma^2}\nabla_\theta \mu = \frac{f^{(l,k)}(x;\theta) - f^{(l,k)}(x;\theta_0)}{\sigma^2}\nabla_\theta f^{(l,k)}(x;\theta) \tag{24}$$

More generally, using the score function identity for exponential families, we can write:

$$\nabla_\theta D_{KL}(p\|q) = \mathbb{E}_p\left[\nabla_\theta \log q\right] = \left(1 + \log\frac{q}{p}\right)\nabla_\theta q \tag{25}$$

Summing over all features in $\mathcal{S}$ and taking expectations completes the proof. □

C Proof of Theorem 12 (Safety Bound)

Theorem (Restated)**.** *Let θ_0 be an aligned model and $\theta^{(T)}$ be the model after T steps of CAE training. Under Assumption 10 (Conditions 1 and 2), if $\mathcal{L}_{anchor}(\theta^{(T)}) \le \epsilon$, then:*

$$|\mathbb{E}_{x\sim\mathcal{D}_{harm}}\left[\pi_{\theta^{(T)}}(y_{refuse}|x) - \pi_{\theta_0}(y_{refuse}|x)\right]| \le C\sqrt{\epsilon} \tag{26}$$

where C depends on B, L_f, and $|\mathcal{S}|$. Note that this proof does not require the transcoder to be differentiable (Condition 3).

Proof. The proof proceeds in three steps.

Step 1: From KL to Total Variation.

By Pinsker's inequality, for any two distributions P and Q:

$$\|P - Q\|_{\text{TV}} \leq \sqrt{\frac{1}{2} D_{KL}(P \| Q)} \tag{27}$$

Applying this to the circuit activation distributions:

$$\|P_{\theta_0}^{\mathcal{S}} - P_{\theta^{(T)}}^{\mathcal{S}}\|_{\text{TV}} \leq \sqrt{\frac{1}{2} D_{KL}(P_{\theta_0}^{\mathcal{S}} \| P_{\theta^{(T)}}^{\mathcal{S}})} \leq \sqrt{\frac{\epsilon}{2}} \tag{28}$$

Step 2: Lipschitz Property of Refusal Probability.

We now establish that the refusal probability is Lipschitz in the safety circuit activations. By Definition 6, the safety circuit $\mathcal{S}$ consists of features with high attribution scores for the refusal output. This means:

$$\pi_\theta(y_{\text{refuse}}|x) \approx g(f_{\mathcal{S}}(x; \theta)) \tag{29}$$

for some function g that depends primarily on the safety features.

By Assumption 10(2), the feature map is L_f-Lipschitz. Furthermore, the output probability is a softmax function, which is Lipschitz with constant at most 1 in its inputs. Therefore, there exists a constant L_g such that:

$$|\pi_\theta(y_{\text{refuse}}|x) - \pi_{\theta'}(y_{\text{refuse}}|x)| \leq L_g \|f_{\mathcal{S}}(x; \theta) - f_{\mathcal{S}}(x; \theta')\| \tag{30}$$

Step 3: Combining the Bounds.

By the definition of total variation distance:

$$\begin{align}
&|\mathbb{E}_x [\pi_{\theta^{(T)}}(y_{\text{refuse}}|x)] - \mathbb{E}_x [\pi_{\theta_0}(y_{\text{refuse}}|x)]| \tag{31}\\
&\leq \mathbb{E}_x [|\pi_{\theta^{(T)}}(y_{\text{refuse}}|x) - \pi_{\theta_0}(y_{\text{refuse}}|x)|] \tag{32}\\
&\leq L_g \mathbb{E}_x \left[\|f_{\mathcal{S}}(x; \theta^{(T)}) - f_{\mathcal{S}}(x; \theta_0)\|\right] \tag{33}\\
&\leq L_g \sqrt{|\mathcal{S}|} \cdot \|P_{\theta_0}^{\mathcal{S}} - P_{\theta^{(T)}}^{\mathcal{S}}\|_{\text{TV}} \cdot B \tag{34}\\
&\leq L_g \sqrt{|\mathcal{S}|} \cdot B \cdot \sqrt{\frac{\epsilon}{2}} \tag{35}
\end{align}$$

where the third inequality uses Cauchy-Schwarz and the bounded activation assumption.

Setting $C = L_g \sqrt{|\mathcal{S}|} \cdot B / \sqrt{2}$ completes the proof. □

D Additional Theoretical Results

D.1 Convergence Analysis

We provide a convergence guarantee for the CAE algorithm under standard assumptions.

Assumption 16 (Smoothness)**.** The losses $\mathcal{L}_{\text{evol}}$ and $\mathcal{L}_{\text{circuit}}$ are β-smooth, i.e., their gradients are β-Lipschitz:

$$\|\nabla \mathcal{L}(\theta) - \nabla \mathcal{L}(\theta')\| \leq \beta \|\theta - \theta'\| \tag{36}$$

Theorem 17 (Convergence Rate)**.** *Under Assumptions 4.5 and 16, running Algorithm 1 with learning rate $\eta = \frac{1}{\beta(1+\lambda)}$ for T iterations yields:*

$$\min_{t \leq T} \|\nabla \mathcal{L}_{CAE}(\theta^{(t)})\|^2 \leq \frac{2\beta(1+\lambda)(\mathcal{L}_{CAE}(\theta^{(0)}) - \mathcal{L}_{CAE}^*)}{T} \tag{37}$$

where $\mathcal{L}_{CAE}^*$ is the optimal value.

Proof. By β-smoothness of $\mathcal{L}_{\text{CAE}} = \mathcal{L}_{\text{evol}} - \lambda\mathcal{L}_{\text{circuit}}$:

$$\mathcal{L}_{\text{CAE}}(\theta^{(t+1)}) \leq \mathcal{L}_{\text{CAE}}(\theta^{(t)}) + \langle \nabla\mathcal{L}_{\text{CAE}}(\theta^{(t)}), \theta^{(t+1)} - \theta^{(t)} \rangle \tag{38}$$

$$+ \frac{\beta(1+\lambda)}{2}\|\theta^{(t+1)} - \theta^{(t)}\|^2 \tag{39}$$

Substituting $\theta^{(t+1)} - \theta^{(t)} = \eta\nabla\mathcal{L}_{\text{CAE}}(\theta^{(t)})$:

$$\mathcal{L}_{\text{CAE}}(\theta^{(t+1)}) \leq \mathcal{L}_{\text{CAE}}(\theta^{(t)}) - \eta\|\nabla\mathcal{L}_{\text{CAE}}(\theta^{(t)})\|^2 + \frac{\beta(1+\lambda)\eta^2}{2}\|\nabla\mathcal{L}_{\text{CAE}}(\theta^{(t)})\|^2 \tag{40}$$

$$= \mathcal{L}_{\text{CAE}}(\theta^{(t)}) - \eta\left(1 - \frac{\beta(1+\lambda)\eta}{2}\right)\|\nabla\mathcal{L}_{\text{CAE}}(\theta^{(t)})\|^2 \tag{41}$$

With $\eta = \frac{1}{\beta(1+\lambda)}$:

$$\mathcal{L}_{\text{CAE}}(\theta^{(t+1)}) \leq \mathcal{L}_{\text{CAE}}(\theta^{(t)}) - \frac{1}{2\beta(1+\lambda)}\|\nabla\mathcal{L}_{\text{CAE}}(\theta^{(t)})\|^2 \tag{42}$$

Summing from $t = 0$ to $T - 1$ and rearranging:

$$\sum_{t=0}^{T-1}\|\nabla\mathcal{L}_{\text{CAE}}(\theta^{(t)})\|^2 \leq 2\beta(1+\lambda)(\mathcal{L}_{\text{CAE}}(\theta^{(0)}) - \mathcal{L}_{\text{CAE}}(\theta^{(T)})) \tag{43}$$

Since $\mathcal{L}_{\text{CAE}}(\theta^{(T)}) \geq \mathcal{L}^*_{\text{CAE}}$:

$$\min_{t\leq T}\|\nabla\mathcal{L}_{\text{CAE}}(\theta^{(t)})\|^2 \leq \frac{1}{T}\sum_{t=0}^{T-1}\|\nabla\mathcal{L}_{\text{CAE}}(\theta^{(t)})\|^2 \leq \frac{2\beta(1+\lambda)(\mathcal{L}_{\text{CAE}}(\theta^{(0)}) - \mathcal{L}^*_{\text{CAE}})}{T} \tag{44}$$

□

D.2 Sample Complexity for Circuit Identification

We analyze the number of samples needed to reliably identify the safety circuit.

Theorem 18 (Circuit Identification Sample Complexity)**.** *Let $\mathcal{S}^*$ be the true safety circuit and $\hat{\mathcal{S}}_n$ be the circuit estimated from n samples. Under Assumption 10, with probability at least $1 - \delta$:*

$$|\hat{\mathcal{S}}_n \triangle \mathcal{S}^*| \leq O\left(\frac{LK\log(LK/\delta)}{n\gamma^2}\right) \tag{45}$$

where $\triangle$ denotes symmetric difference and γ is the attribution threshold from Definition 6.

Proof. The attribution score for each feature is estimated as:

$$\hat{\alpha}^{(l,k)} = \frac{1}{n}\sum_{i=1}^{n}\alpha^{(l,k)}_{y_{\text{refuse}}}(x_i;\theta_0) \tag{46}$$

By Hoeffding's inequality, for each feature:

$$\mathbb{P}\left(|\hat{\alpha}^{(l,k)} - \alpha^{(l,k)}| > \epsilon\right) \leq 2\exp\left(-\frac{2n\epsilon^2}{B^2}\right) \tag{47}$$

where B bounds the attribution scores (Assumption 10(1)).

Taking a union bound over all LK features and setting $\epsilon = \gamma/2$:

$$\mathbb{P}\left(\exists(l,k) : |\hat{\alpha}^{(l,k)} - \alpha^{(l,k)}| > \gamma/2\right) \leq 2LK\exp\left(-\frac{n\gamma^2}{2B^2}\right) \tag{48}$$

For this probability to be at most δ, we need:

$$n \geq \frac{2B^2}{\gamma^2}\log\left(\frac{2LK}{\delta}\right) \tag{49}$$

When all estimation errors are at most $\gamma/2$, any feature with true attribution $\geq \gamma$ will have estimated attribution $\geq \gamma/2$, and any feature with true attribution $< \gamma/2$ will have estimated attribution $< \gamma$. This bounds the symmetric difference. □

D.3 Robustness to Circuit Misspecification

We analyze the effect of errors in circuit identification.

Theorem 19 (Robustness)**.** *Let $\mathcal{S}$ be the identified circuit and $\mathcal{S}^*$ be the true safety circuit. If $|\mathcal{S}\triangle\mathcal{S}^*| \leq \epsilon_{\mathcal{S}}|\mathcal{S}^*|$ for some $\epsilon_{\mathcal{S}} \in [0, 1)$, then the safety bound (Theorem 12) degrades gracefully:*

$$|\pi_{\theta^{(T)}}(y_{\textit{refuse}}|x) - \pi_{\theta_0}(y_{\textit{refuse}}|x)| \leq C\sqrt{\epsilon} + C'\epsilon_{\mathcal{S}} \tag{50}$$

where C' depends on the maximum attribution score of misspecified features.

Proof. Decompose the safety circuit activation difference:

$$\|f_{\mathcal{S}^*}(\theta^{(T)}) - f_{\mathcal{S}^*}(\theta_0)\| \leq \|f_{\mathcal{S}}(\theta^{(T)}) - f_{\mathcal{S}}(\theta_0)\| \tag{51}$$

$$+ \|f_{\mathcal{S}^*\setminus\mathcal{S}}(\theta^{(T)}) - f_{\mathcal{S}^*\setminus\mathcal{S}}(\theta_0)\| \tag{52}$$

$$+ \|f_{\mathcal{S}\setminus\mathcal{S}^*}(\theta^{(T)}) - f_{\mathcal{S}\setminus\mathcal{S}^*}(\theta_0)\| \tag{53}$$

The first term is bounded by the circuit constraint: $O(\sqrt{\epsilon})$.

The second term represents missed safety features (false negatives). These are unconstrained, but by the definition of $\mathcal{S}^*$, their total attribution is at most $\gamma|\mathcal{S}^* \setminus \mathcal{S}| \leq \gamma\epsilon_{\mathcal{S}}|\mathcal{S}^*|$.

The third term represents incorrectly included features (false positives). Constraining these does not hurt safety, only potentially capability.

Combining via the Lipschitz property of refusal probability completes the proof. □

E Scalability to Larger Models and Longer Evolution Horizons

To demonstrate that Circuit-Anchored Evolution (CAE) is not limited to smaller models or short evolutionary windows, we conduct extended experiments addressing both model scale and evolution duration.

Scaling to Larger Models. We apply CAE to Gemma-2-9B-IT, evolving it for 500 steps. As shown in Table 3, CAE preserves safety almost perfectly (98.47%) with no significant capability loss compared to unconstrained evolution. This demonstrates that our method does not trade off capability for safety, even as model capacity scales up.

Table 3: Performance on larger model (Gemma-2-9B-IT, 500 steps).

Method	**Safety Score (%)**	**Capability Score (%)**
Unconstrained Evolution	69.85	64.72
Evolution w/ CAE (Ours)	**98.47**	64.58

Durability over Longer Evolution. Standard self-evolution typically converges or is stopped around 500 steps. To stress-test the durability of our anchoring mechanism, we push the evolution of Gemma-2-2B-IT to an extreme horizon of 5,000 steps.

Table 4: Performance under extreme evolution horizon (Gemma-2-2B-IT, 5000 steps).

Method	**Safety Score (%)**	**Capability Score (%)**
Unconstrained Evolution	61.24	45.82
Evolution w/ CAE (Ours)	**95.81**	45.37

As shown in Table 4, unconstrained evolution exhibits further safety degradation over longer horizons (dropping to 61.24%), underscoring the severity of the misevolution problem. In stark contrast, CAE remains remarkably stable even at 5,000 steps, maintaining a 95.81% safety score. This suggests that the circuit anchoring mechanism provides durable, long-term protection against safety degradation, effectively preventing the model from drifting into harmful subspaces regardless of the evolution length.